The Kuramoto Model and Neural Synchronization

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Abstract

The following simulations will introduce adjacency matrices for various undirected networks and a phase frustration parameter into the original Kuramoto equation in order to better represent synchronization dynamics that occur in neural networks. Specifically, the Kuramoto model be used in order to evaluate the effect of phase frustration for a completely connected network and a series of random modulator networks. Furthermore, these simulations will be used in order to investigate the combined effect of phase frustration and modularity on order parameter as a function of coupling strength.

1 Introduction

1.1 The Original Kuramoto Model

The Kuramoto model (derived by Yoshiki Kuramoto in 1975) is mathematical model of collective synchronization that allows for simulating long-term dynamics that underlie nonlinear synchronization in a system of coupled oscillators [2] [3]. This model is comprised of a series first order differential equations that describe changes in phase dynamics of a given oscillator (θ_i) on the unit circle over time due to coupling interactions with other oscillators in a given system. The governing equation of these differential equations is defined as

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i) \tag{1}$$

where ω_i, K, and N denote the internal frequency of the i^{th} oscillator, the coupling strength, and the number of oscillators in the system respectively with the term $\sin(\theta_j - \theta_i)$ characterized as the interaction function between θi and θj [3].

1.2 Order Parameter

In addition to equation (1), Kuramoto's analysis included deriving a parameter that described synchronization of the entire group of oscillators over time [3]. In order to simplify this analysis, Kuramoto assumed that the distribution of the internal frequencies of the system of oscillators $g(\omega)$ was unimodal and symmetric about an average internal frequency ω_0 [3]. This assumption allowed for Kuramoto to treat the system of coupled oscillators as a mean-field system [3]. As a result of his analysis, Kuramoto determined that the centroid complex number (R) of a group of oscillators served as as a measure of phase synchronization [3]. Now known as the order parameter, this centroid value is given by

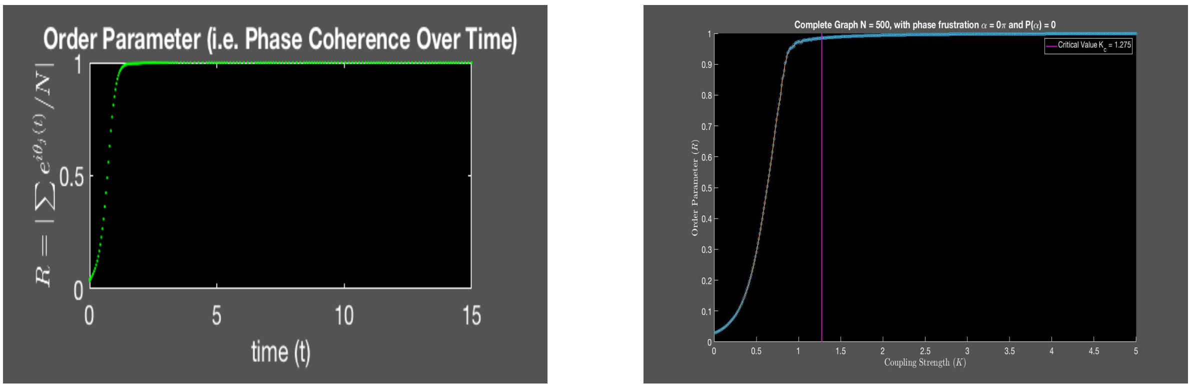
$$R = r(t)\mathrm{e}^{i\psi(t)} = \frac{1}{N}\sum_{j=1}^{N} \mathrm{e}^{i\theta_j} \tag{2}$$

where $r(t)$ and $\psi(t)$ denote the average phase coherence and average phase respectively, as a function of time.

The range defined for the order parameter lies between $(0,1)$, where $R = 0$ indicates no synchronization and $R = 1$ indicates full synchronization. A system of oscillators is said to be in a incoherent state when $R \approx 0$ and in a coherent state when $R \approx 1$ [3]. As shown in Figure (1), if the coupling strength surpasses some critical value (K_c), then a positive feedback between coupling and average phase coherence occurs that results in a rapid transition from the incoherent to the coherent state. After this transition occurs, the order parameter saturates in this coherent state $(0 << R_\infty < 1)$, with occasional $O(N^{1/2})$ fluctuations [3] [3]. Another characteristic of the order parameter demonstrated in Figure (1) is that there is a sharp increase in (R_∞) from incoherence $(0 < R_\infty << 1)$ to coherence $(0 << R_\infty < 1)$ near some critical value K_c [3]. For his analysis, Kuramoto defined the critical value as

$$K_c = \frac{2}{\pi(g\omega_0)} \tag{3}$$

where $g(\omega_0)$ denotes the average internal frequency of a group of oscillators.



(a) $R(t)$ for $N = 500$ Oscillators at $K = 7$ Coupling Strength

(b) Critical Coupling Strength (K_c) and R_∞ as a Function of Coupling Strength (K)

Figure 1: Order Parameter for the Original Kuramoto Model

2 Spatial embedding and Phase Frustration

Due to the simplicity of the Kuramoto model, it has been widely used as a foundation for studying collective synchronization of various systems that occur in nature. However, synchronization that occurs in a majority of these systems are far more complex than that described by the original Kuramoto model. Introducing new parameters into the Kuramoto equation easily overcomes this limitation and allows for the generation of more complex models that are comparable with some specific system in nature. The simulations generated below utilize this method in order to generate a model that better describes synchronization in neural systems using a coupling matrix and phase frustration parameter.

2.1 Coupling Matrix

Kuromoto's original model described a system of oscillators with "all-to-all" coupling, meaning that all of the oscillators in the system were coupled to one another [1]. This feature can be interpreted using concepts from graph theory, as an undirected complete, k-regular graph, K_N comprised of N nodes and $n(n-1)/2$ edges. This analogy allows the system of oscillators to be defined in terms of a graph where each node in K_N represents an oscillator and each edge exists between a unique node pair and represents a coupling interaction between oscillator i and oscillator j.

Using these concepts from graph theory allows for the direct manipulation of coupling between oscillator pairs. This is achieved by introducing a coupling adjacency matrix ($A_{i,j}$) for some arbitrary graph into the original Kuramoto equation. The resulting equation is given by

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N} A_{i,j}\sin(\theta_j - \theta_i) \tag{4}$$

which can be thought of as the change in phase dynamics over time for a given oscillator in a network of oscillators whose coupling topology is characterized by the coupling adjacency matrix $A_{i,j}$.

2.2 Phase Frustration

Implementing a phase frustration parameter ($\alpha_{i,j}$) in to the Kuramoto model can be used to prevent complete synchronization from occurring [1]. Specifically, it skews the the interaction function for each coupled oscillator in the system and can take on values between $(-\pi, \pi)$. The Kuramoto model with the phase frustration parameter (with the coupling matrix included) is given by

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^{N} A_{i,j} \sin(\theta_j - \theta_i - \alpha_{i,j}) \tag{5}$$

3 Simulations

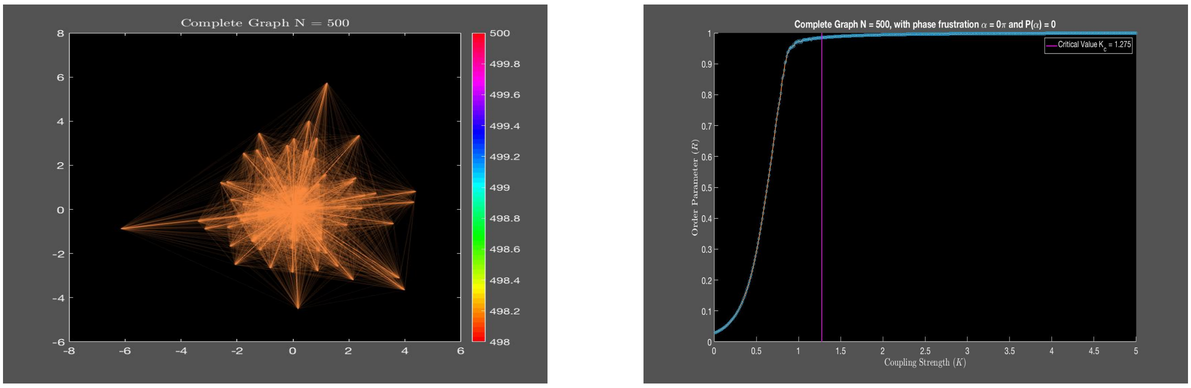
Due to the fact that the nervous system demonstrated synchronization from the cellular to the anatomical level, using a controlled systems approach with the Kuramoto model has great potential in enhancing our current understanding of the nervous system. The following simulations will depict the effects of two fundamental characteristics of neural networks, modularity and time delay between cell interactions, on collective synchronization [1]. Modularity will be accounted for using a coupling matrix that corresponds with a random modular graph and time delays will be accounted for using positive/negative phase frustration parameters [1].

3.1 Procedure

The Kuramoto model was implemented in Matlab into a system of $N = 500$ oscillators with initial phase conditions normally and randomly distributed between $(0, 2\pi)$ and intrinsic frequencies uniformly and randomly distributed between $(0, 1)$. These distributions were executed using the beta distribution and the appropriate parameters in order to generate the uniform and normal distributions. Simulations for the order parameter as a function of coupling strength were created for a K_N network and a random modular network. Additionally, each simulation for the order parameter as a function of coupling strength includes a vertical line representing the critical coupling strength K_c as defined by equation 3. It is important to note that as these coupling strengths do not necessarily apply to simulations that include phase frustration, however these values are included in the figure for comparison of the the critical coupling strength of the system under conditions assumed in the original Kuramoto model.

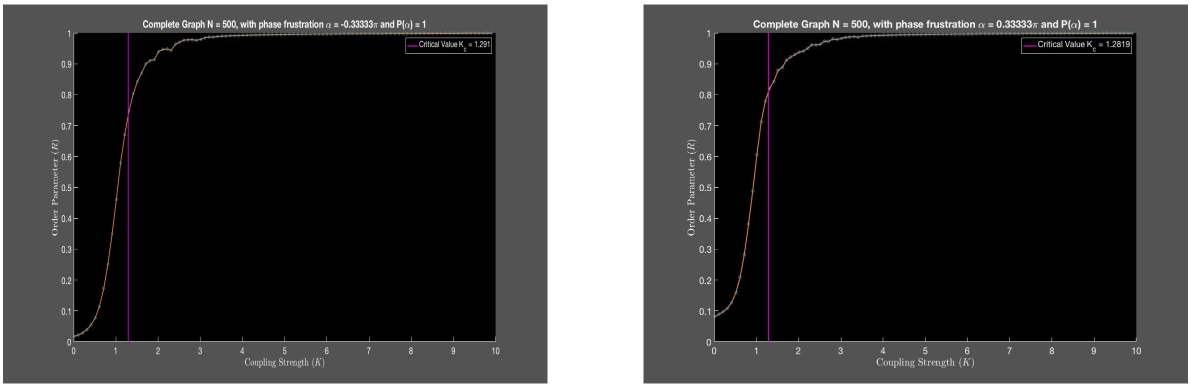
3.2 K_N Network

As mentioned in section two, the K_n network depicts the "all-to-all" coupling condition assumed in the original Kuramoto model (Figure 2). The order parameter for this network was analyzed as a function of coupling strength under six different phase frustration conditions ($\alpha_{i,j} = \pm\frac{1}{3}\pi, \pm\frac{1}{2}\pi$, and $\pm\pi$) (Figure 3). All order parameter simulations depict the order parameter of the system analyzed at $tmax = 15$.

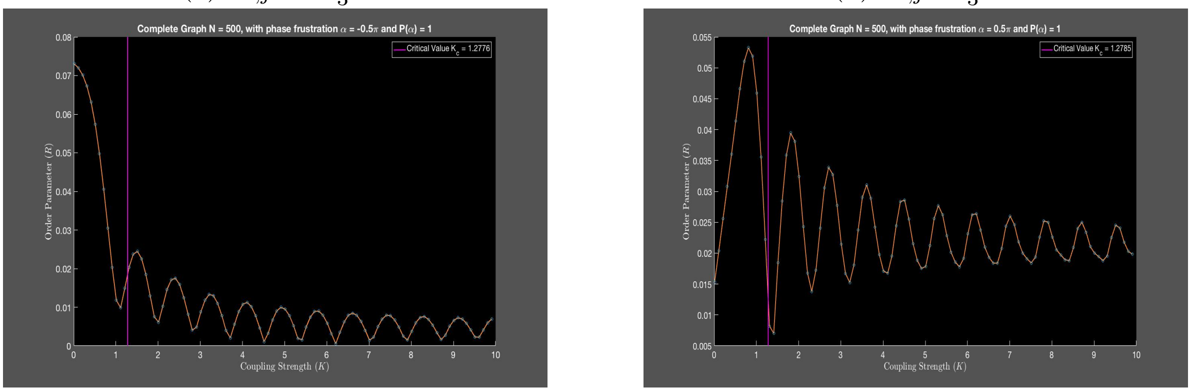


(a) (b)

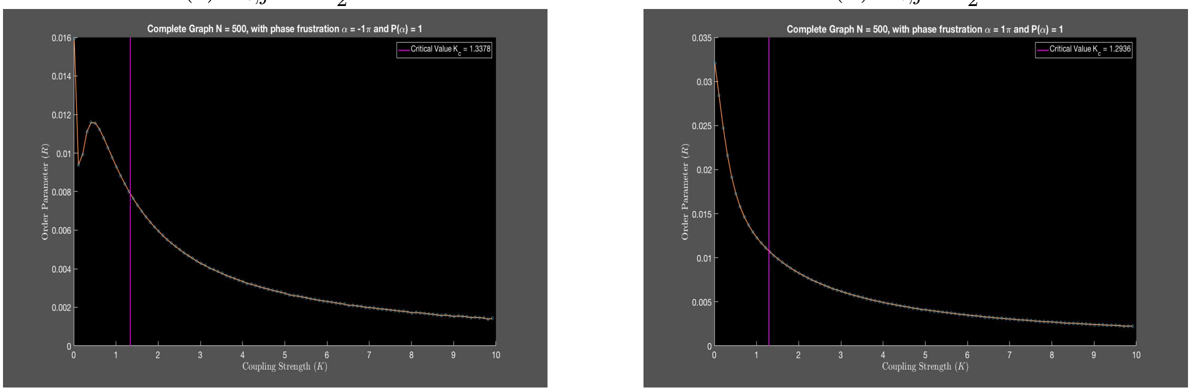
Figure 2: Completely Coupled (K_N) Network and Order Parameter



(a) $\alpha_{i,j} = -\frac{1}{3}\pi$ (b) $\alpha_{i,j} = \frac{1}{3}\pi$



(c) $\alpha_{i,j} = -\frac{1}{2}\pi$ (d) $\alpha_{i,j} = \frac{1}{2}\pi$

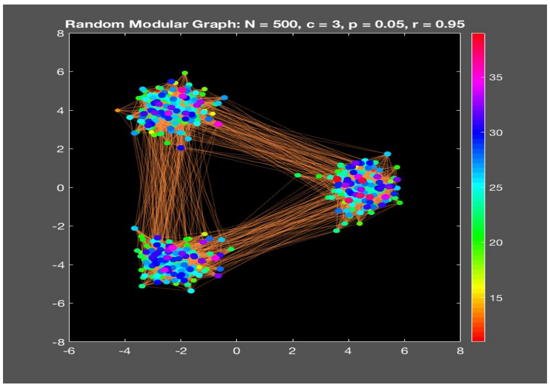


(e) $\alpha_{i,j} = -\pi$ (f) $\alpha_{i,j} = \pi$

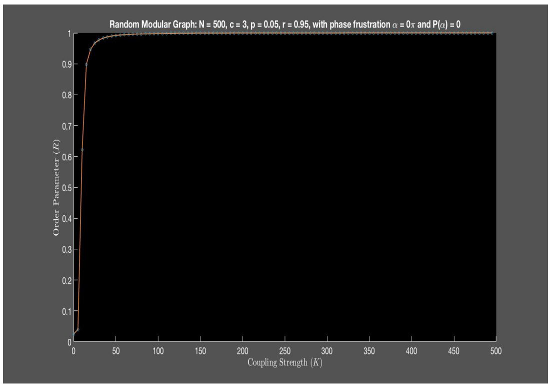
Figure 3: Effect of Phase Frustration on Order Parameter ($R_{t=50}(K)$) for K_N Network

3.3 Random Modular Networks (N, c, p, k)

In order to investigate the effect of modularity (or clusters) on the order parameter as a function of coupling strength, adjacency matrices analogous to random modular networks were generated using Matlab. The parameters associated with these adjacency matrices are defined as the number of oscillators in the network (N), the number of modules/clusters in the network (c), the overall probability of coupling interaction between an arbitrary pair of oscillators (p) and the probability of a coupling interaction between an arbitrary pair of oscillators contained in the same module (0.95). Like the completely coupled network, N was set to 500 for all simulations, and in order to generate distinct module groups, all random modular simulations had parameter values $p = 0.05$ and $k = 0.95$. Order parameter as a function of coupling strength were obtained for clustering values, $c = 3$ (Figure 4), $c = 5$ (Figure 6), and $c = 20$ (Figure 8). Furthermore, for each clustering value, the effect of phase frustration on order parameter as a function of coupling strength was analyzed under the same conditions as those used for the complete graphs (Figures 5, 7, and 9). All order parameter simulations depict the order parameter of the system analyzed at $tmax = 50$.



(a)

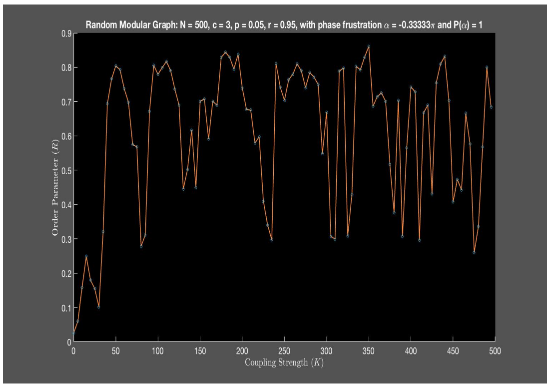


(b)

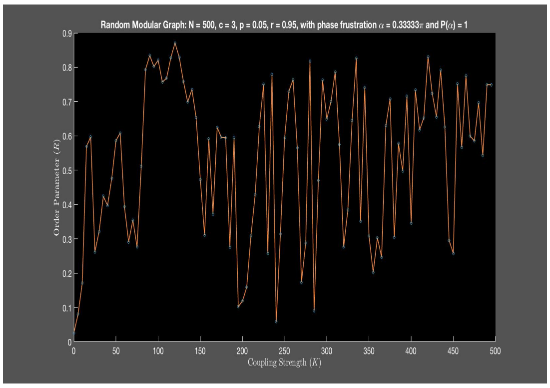
Figure 4: Random Modular Network $(500, 3, 0.05, 0.95)$ and Order Parameter

4 Discussion

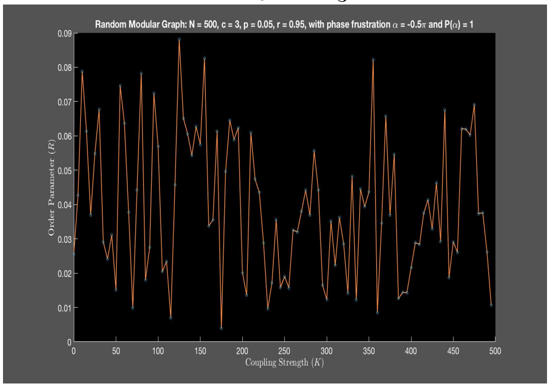
As expected, increasing phase frustration from $\pm\frac{1}{3}\pi$ to $\pm\pi$ increased the reducing effect on order parameter. For the complete graphs, phase frustration at $\pm\frac{1}{3}\pi$ increased the necessary coupling strength value in order for global synchronization to occur (Figure 3a and 3b) compared to the critical coupling strength of a non-frustrated completely coupled network (Figure 2b). Interestingly, as phase frustration was increased to $\pm\frac{1}{2}\pi$ and $\pm\pi$, the order parameter began to decrease as a function of coupling strength. A similar, but much more complex pattern can be seen for the random modular graphs (Figures 5, 7, and 9). However, it is important to note that the dynamics for the order parameter for both the complete and



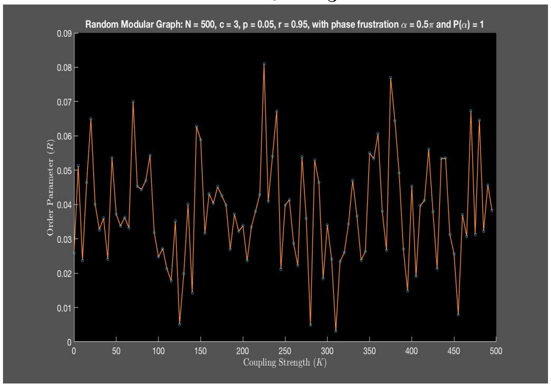
(a) $\alpha_{i,j} = -\frac{1}{3}\pi$



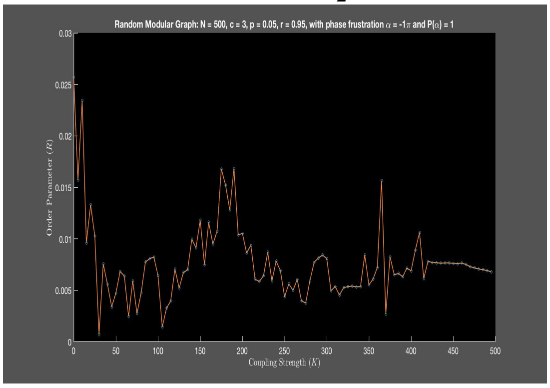
(b) $\alpha_{i,j} = \frac{1}{3}\pi$



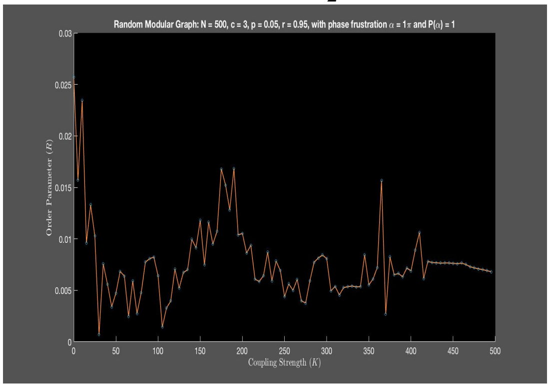
(c) $\alpha_{i,j} = -\frac{1}{2}\pi$



(d) $\alpha_{i,j} = \frac{1}{2}\pi$

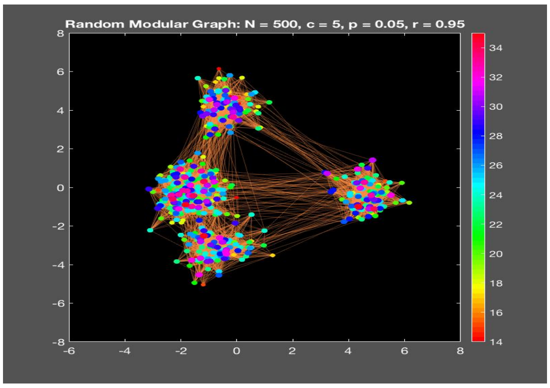


(e) $\alpha_{i,j} = -\pi$

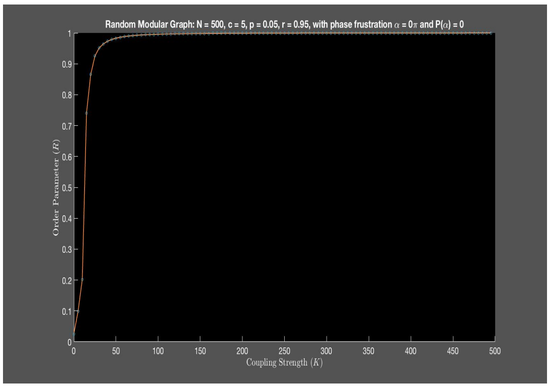


(f) $\alpha_{i,j} = \pi$

Figure 5: Effect of Phase Frustration on Order Parameter ($R_{t=50}(K)$) for Random Modular Networks with 3 Modules

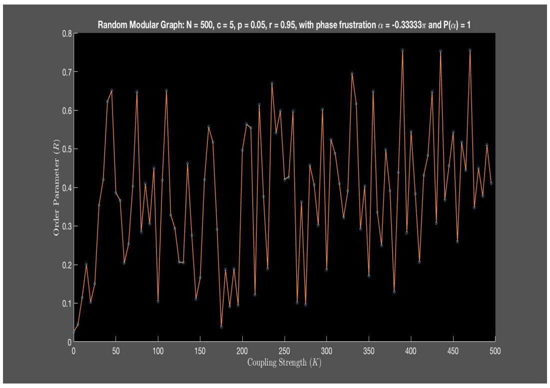


(a)

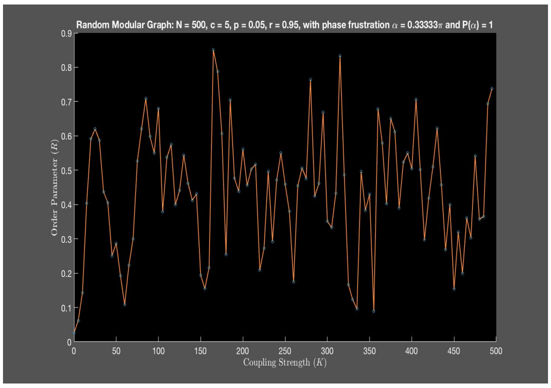


(b)

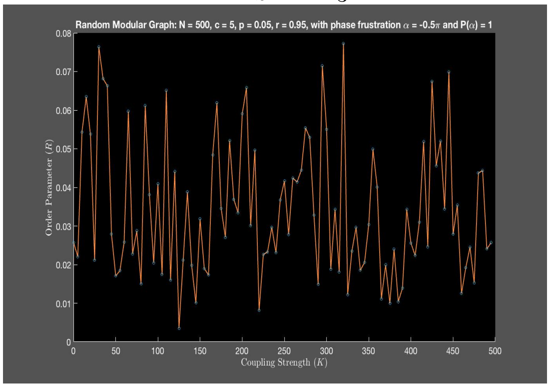
Figure 6: Random Modular Network $(500, 5, 0.05, 0.95)$ and Order Parameter



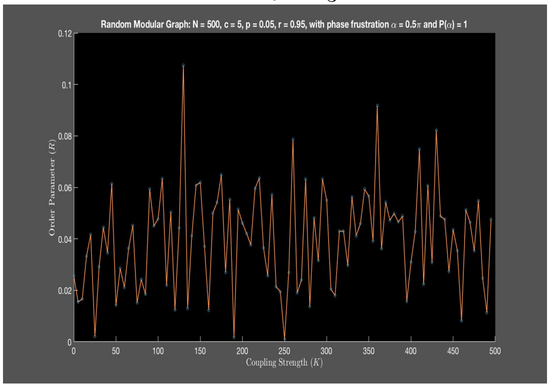
(a) $\alpha_{i,j} = -\frac{1}{3}\pi$



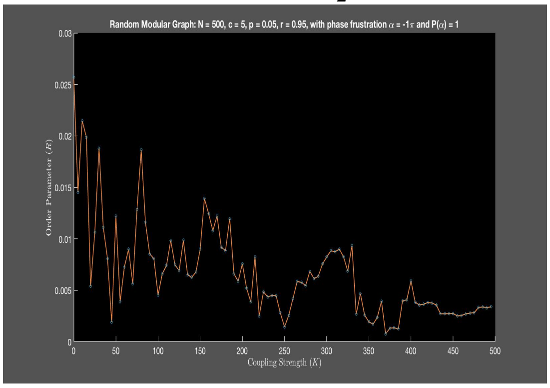
(b) $\alpha_{i,j} = \frac{1}{3}\pi$



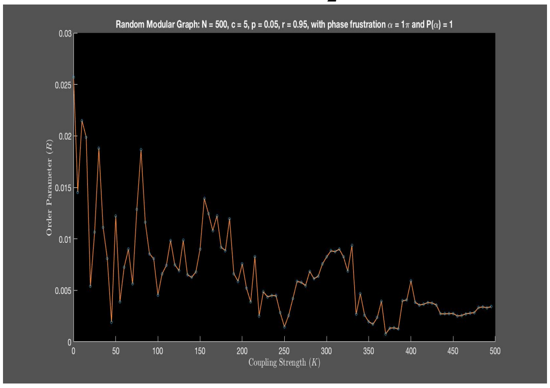
(c) $\alpha_{i,j} = -\frac{1}{2}\pi$



(d) $\alpha_{i,j} = \frac{1}{2}\pi$

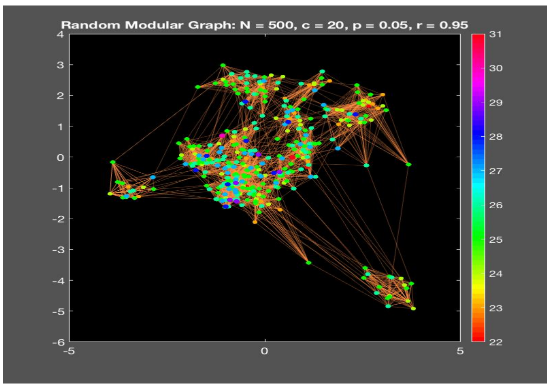


(e) $\alpha_{i,j} = -\pi$

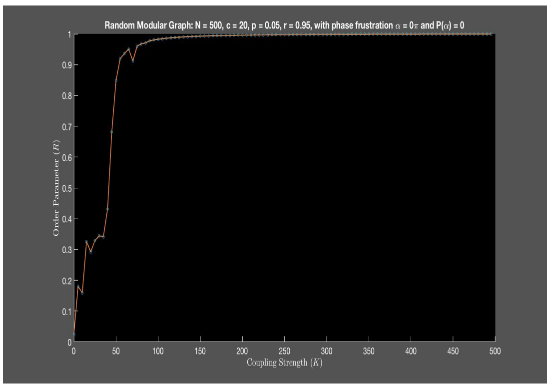


(f) $\alpha_{i,j} = \pi$

Figure 7: Effect of Phase Frustration on Order Parameter ($R_{t=50}(K)$) for Random Modular Networks with 5 Modules

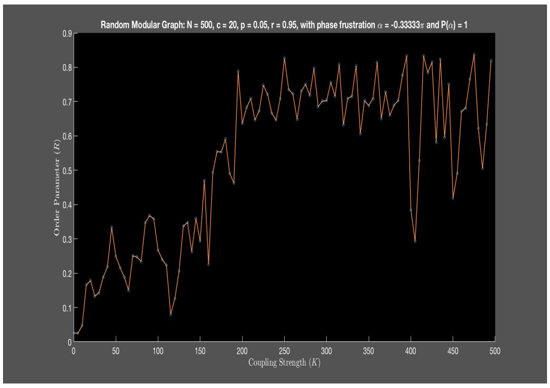


(a)

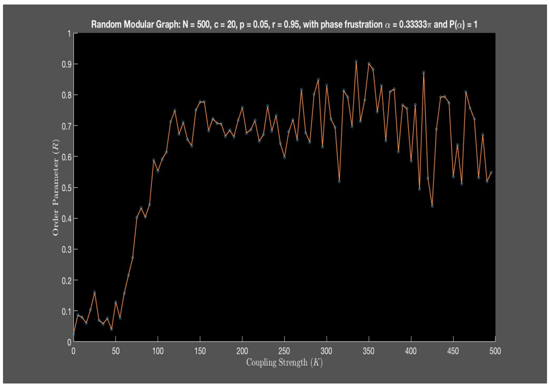


(b)

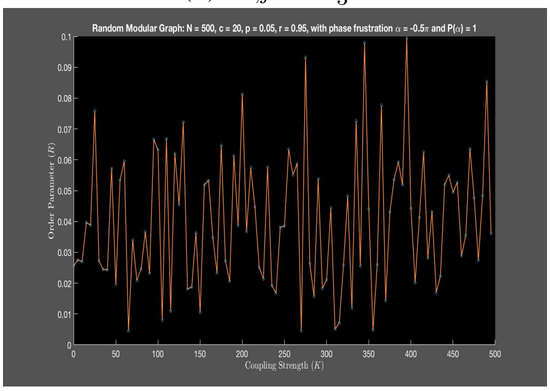
Figure 8: Random Modular Network $(500, 20, 0.05, 0.95)$ and Order Parameter



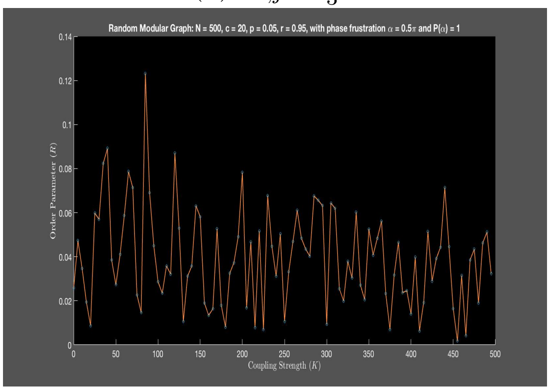
(a) $\alpha_{i,j} = -\frac{1}{3}\pi$



(b) $\alpha_{i,j} = \frac{1}{3}\pi$



(c) $\alpha_{i,j} = -\frac{1}{2}\pi$



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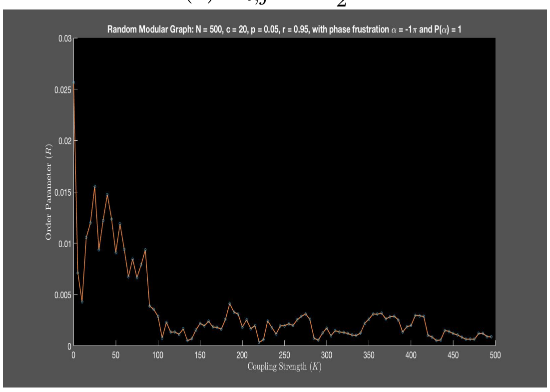
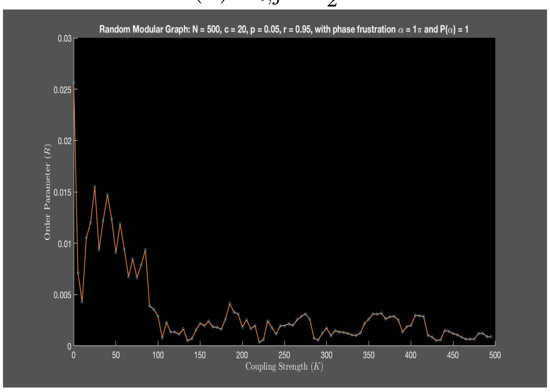


(e) $\alpha_{i,j} = -\pi$



(f) $\alpha_{i,j} = \pi$

Figure 9: Effect of Phase Frustration on Order Parameter ($R_{t=50}(K)$) for Random Modular Networks with 20 Modules

random modular graphs with as a function of time do not saturate at some steady state for phase frustration conditions. More specifically, these values undergo seemingly random oscillations between at some range between 0 and some value less than one. This fact complicates the interpretation of the order parameter at $tmax$ as a function of coupling strength, as they do not represent the order parameter as $tmax \to \infty$. This complication demonstrates that the Kuramoto model rapidly increases in complexity as parameters are added into the system.

An interesting finding regarding modularity is that as the number of modules in the system increases, the oscillations in order parameter as a function of coupling strength seem to follow a pattern comparable to that of the complete graph for various phase frustration values. Since neural networks are known to have high modularity and undergo time delays, this finding may have significance to synchronization patterns in neural networks. Future research should expand on these findings by including additional parameters that can increase comparability between the Kuramoto model and brain networks. Two important feature to include would be variations in postsynaptic neural excitability and synaptic neural inhibition or excitation.

5 Conclusion

The Kuramoto model serves as a very general, but useful mechanism for describing synchronization dynamics that can be applied to a number of synchronization phenomenon that occur in nature [3]. Furthermore, the simplicity of the Kuramoto Model allows for small changes to be made in order to better accommodate mathematical analysis of specific synchronization situations [1]. Specifically, the Kuramoto model has proven to be an extremely useful tool for modeling neuron synchronization and implementation of the model with concepts derived from network theory and has great potential in future research as a means for better understanding neural synchronization dynamics at the cellular and anatomical scales [1].

References

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